

Switching-like Event-triggered Control for Networked Control Systems under Malicious Denial of Service Attacks

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Abstract—This paper investigates the switching-like event-triggered control for networked control systems (NCSs) under the malicious denial of service (DoS) attacks. First, by dividing the DoS attacks into S-interval (DoS-free case) and D-interval (DoS case), a switching-like event-triggered communication scheme (SETC) is well designed to deal with intermittent DoS attacks to improve communication efficiency while keeping the desired control performance. Secondly, by considering the SETC and NCSs into a unified framework, the studied system is transferred into a time-delay system. Then, under the constraint of the number of maximum allowable data dropouts induced by DoS attacks, a stability criterion and a stabilization criterion are derived, which can be used to estimate the event-triggered communication parameters and obtain the security controller gain simultaneously. Moreover, the derived stabilization criterion can also provide a trade-off to balance communication efficiency and H_∞ control performance. At last, a networked invert pendulum on a cart is conducted to show the effectiveness of the proposed method.

Index Terms—Networked control systems, Denial of service attacks, Switching-like event-triggered communication, Security control.

I. INTRODUCTION

Networked control systems (NCSs) have received more attention in the past few decades because of their widespread applications, like automobiles, multi-agent systems, smart grids and so on [1], [2]. However, by characterizing the deep integration of cyberspace and physical systems, there are some inherent challenges on the NCSs due to limitations of time delay, packet dropouts, quantization error and communication constraints. Up to now, many results are presented in modeling, stability analysis and controller design of NCSs [3]. It is worth noting that recent incidents such as ‘Stuxnet’, ‘Flame’ show that malicious attacks on cyberspace would make another real threat to NCSs, this stimulates broad interest to the control community [4], [5].

Generally, cyber attacks can be divided into denial of service (DoS) attacks and deception attacks [6], [7], DoS is perhaps one of the most detrimental attacks by blocking the packets delivery between measurement and control channels. Different from the traditional time delay and packet dropouts,

DoS attacks can cause a more serious case which would arouse the degradation of control performance, even instability [8], [9]. Therefore, different from traditional IT protection strategies, it is necessary to develop some security strategies from the perspective of control-theoretic framework. In the literature, there are some methods to describe the behavior of DoS attacks, such as periodic DoS attacks [6], stochastic DoS attacks [10], [11] and arbitrary DoS attacks [12], [13], [14]. Correspondingly, some effective methods are developed to analyze and control of networked control systems under DoS attacks [15], [16], [17]. A real attacker will launch their attacks without following any determined rules such as periodic features and specific probability distributions, and their properties are rarely known ahead of time. This leads to many difficulties in the analysis and synthesis of NCSs under DoS attacks.

Recently, a so-called event-triggered control (ETC) scheme has been efficiently applied to deal with the scenario that the communication channels with limited bandwidth [18], [19], [20]. Generally, ETC scheme can be classified into continuous event-based scheme [21], [22] and discrete event-based scheme [19], [23]. However, a common assumption on the above work is that all event-triggered packets should be successfully transmitted to the sequential nodes. Most communication networks are open and vulnerable to DoS attacks. In this case, DoS attacks would disrupt the pre-designed ‘event’ and even make the system instability. Therefore, the above ETC schemes no longer valid again and should be re-evaluated and re-designed under DoS attacks.

More recently, there are some novel works on event-triggered NCSs with DoS attacks, i.e., under the ETC framework and with the pre-given security controller gain, periodic DoS attacks are considered in [24], [25] and arbitrary DoS attacked besides energy/frequency constraints are investigated in [12], [16]. A resilient event-triggered communication scheme is proposed in [16], which can estimate the allowable number of continuous data losses induced by DoS attacks. However, a larger number of packets must be transmitted since smaller event-triggered communication parameter is always used in the communication. By assuming a period DoS attack, a novel security controller design method is proposed in [25]. Compared with traditional ETC without considering DoS attacks, more event-triggered packets should be generated to deal with possible packet dropouts induced by DoS attacks. In a practical communication network, the DoS attacks are sneaking and intermittent with the limitations of attack fre-

This work was supported in part by the National Natural Science Foundation of China under Grants 61833011, 61673255, 61633016, the 111 Project under Grant D18003, and the Outstanding Academic Leader Project of Shanghai Science and Technology Commission under Grant 18XD1401600.

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quency and attack energy to hide the malicious attacks. If above-mentioned properties are well considered in the design of communication scheme and security controller, higher communication efficiency can be expected. These motivate the current study.

This paper attempts to establish a switching-like event-triggered control framework for networked control systems to improve communication efficiency while keeping the desired control performance. The constrained attack frequency and attack energy are well considered in the analysis and synthesis of NCSs with DoS attacks. Both the event-triggered communication parameter and the security controller gains can be simultaneously obtained based on the derived criteria. The main contributions of this paper can be summarized as

- 1) By fully use of the DoS attack properties, a switching-like event-triggered communication scheme is well designed to improve the communication efficiency. Compared with resilient event-triggered communication scheme in [16], the proposed communication scheme can reduce the number of transmitted packets by actively choosing the appropriate event-triggered communication scheme;
- 2) A time delay model is constructed to describe the studied system under DoS attacks, which is convenient for us to consider the design of event-triggered communication scheme and security controller in a unified framework; and
- 3) By exploiting the derived stabilization criteria, the event-triggered communication parameters and security controller gain can be obtained at the same time under consideration of energy-limited DoS attacks. Moreover, the proposed method provides a trade-off to balance communication efficiency and H_∞ control performance.

The rest of this paper is organized as follows. Section II gives some preliminaries of event-triggered control framework under DoS attacks as well as some definitions about attack constraints. Section III provides a switching-like event-triggered communication scheme under consideration of DoS attacks. In section IV, we present our main results which include the asymptotically stability criterion and the security controller design method for NCSs with energy-limited DoS attacks. Then simulation results are used to verify the proposed method in Section V and the last section concludes this paper. *Notation:* In this paper, $\mathbb{R}^n$ denotes the n -dimensional Euclidean space and $\mathcal{L}_2$ is Euclidean norm represented by $|\star|$. $R_{(>0)}$ is the set of (positive) reals. The superscript “ T ” refers to matrix transposition. Matrix $X > 0$ ($X \geq 0$) implies that X is a positive definite (semi-definite) symmetric matrix. The symbol $*$ gives the symmetric term in a symmetric matrix.

II. PROBLEM FORMULATION

In this section, the framework of event-triggered NCSs with DoS attacks is described in II-A, the description of DoS attacks with limited attack energy is given in II-B.

A. Event-triggered control framework under DoS attacks

Assume that there is a communication network between the sensor and the controller, an event-triggered communication

scheme is used to decide whether or not to transmit the current sampled-data for saving limited communication resource, and the event-triggered packets may be lost due to malicious DoS attacks. The framework of event-triggered NCSs under DoS attacks is schematically described in Fig. 1.

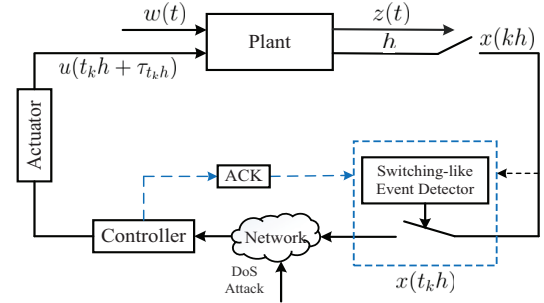


Fig. 1. Event-triggered control framework under DoS attacks

The dynamics of the plant to be controlled can be described by the following differential equations

$$\begin{cases} \dot{x}(t) = Ax(t) + Bu(t) + B_w w(t) \\ z(t) = Cx(t) + Du(t) \end{cases} \quad (1)$$

where $x(t) \in \mathbb{R}^n$, $u(t) \in \mathbb{R}^m$ and $z(t) \in \mathbb{R}^p$ are state vector, input vector, and regulated output vector, respectively. $w(t) \in \mathcal{L}_2[0, \infty)$ is a class of physical disturbances. A, B, B_w, C, D are constant matrices with compatible dimensions. The initial condition of the system (1) is given by $x(t_0) = x_0$.

Suppose that the sensor is time-triggered with sampling period h . Then the sampling sequence is described by the set $\mathcal{S}_1 = \{0, 1h, 2h, \dots, kh, \dots\}, k \in \mathbb{N}$. By collecting these sampling data, similar to [16], one can design an event-triggered communication scheme as

$$s_{k+1} = s_k + \inf\{\ell h | e^T(s_k + \ell h) \Phi e(s_k + \ell h) \geq \delta_* x(s_k)^T \Phi x(s_k)\} \quad (2)$$

where s_k is current transmitted sampling instant and s_{k+1} is future transmitted instant, $e(s_k + \ell h) = x(s_k + \ell h) - x(s_k)$ is the error between the current state $x(s_k + \ell h)$ and the latest transmitted state $x(s_k)$, $\ell = 1, 2, \dots$, δ_* and Φ need to be designed appropriately.

Set $\mathcal{S}_2 = \{s_0, s_1, s_2, \dots, s_k, \dots\}$, which is the set of event-triggered packets needed to be transmitted by (2). Clearly, $\mathcal{S}_2 \subseteq \mathcal{S}_1$. Then the actuators will implement their control actions with these successfully transmitted sampled data in a ZOH-style. The control action can be given as follows

$$u(t) = Kx(s_k), t \in [s_k + \tau_{s_k}, s_{k+1} + \tau_{s_{k+1}}) \quad (3)$$

where τ_{s_k} is the communication induced delay and K is a controller gain to be designed. Then, this leads to the following closed-loop NCS model

$$\begin{cases} \dot{x}(t) = Ax(t) + Bu(t) + B_w w(t) \\ z(t) = Cx(t) + Du(t) \\ u(t) = Kx(s_k), t \in [s_k + \tau_{s_k}, s_{k+1} + \tau_{s_{k+1}}) \end{cases} \quad (4)$$

B. Energy-limited DoS attacks

As described in Fig. 1, DoS attacks can prevent the control updates from being executed at the desired time, i.e., an event-triggered packet is lost in the communication with DoS attacks. Similar to [12], we also denote the sequence of DoS from *off* to *on* as $\{h_n\}$, which represents the communication is possible. A DoS attack interval can be represented as

$$\mathcal{H}_n = \{h_n\} \cup [h_n, h_n + \zeta_n) \quad (5)$$

where $\zeta_n \in R_{\geq 0}$ denotes the time length of DoS attacks and the DoS attack takes the form of a single pulse at time h_n if $\zeta_n = 0$. For the convenience of analysis, the union interval of DoS attacks during the time interval $[0, t)$ is written as

$$\mathcal{D}(0, t) = \bigcup_{n \in \mathbb{N}} \mathcal{H}_n \quad (6)$$

Let $\mathcal{S}_3 = \{t_0, t_1, \dots, t_k, t_{k+1}, \dots\} \subseteq [0, t)$, which denotes the set of successful transmission event-triggered instants during time interval $[0, t)$ with $k = 1, 2, \dots, N$ and $t \geq 0$. From the definitions of $\mathcal{S}_1, \mathcal{S}_2$, and $\mathcal{S}_3$, it is known that $\mathcal{S}_3 \subseteq \mathcal{S}_2 \subseteq \mathcal{S}_1$. In order to characterize such jamming attacks under the event-triggered framework for NCSs, the following definition is used to classify the transmission behaviors for every successful transmission instant.

Definition 1. *There are following two categories for each successful transmission interval $[t_k, t_{k+1})$, $k \in \mathbb{N}$ when the NCSs subjects to DoS attacks*

- *S-interval $\mathcal{I}_{t_k}$: If $[t_k, t_{k+1}) \cap \mathcal{D}(0, t) = \emptyset$, there are no DoS attacks during time interval $[t_k, t_{k+1})$.*
- *D-interval $\bar{\mathcal{I}}_{t_k}$: If $[t_k, t_{k+1}) \cap \mathcal{D}(0, t) \neq \emptyset$, there are DoS attacks during time interval $[t_k, t_{k+1})$.*

From the above Definition, one can see that for any $t > t_0 \geq 0$, the total number of the event-triggered time interval $N_T(t_0, t) = N_D(t_0, t) + N_S(t_0, t)$, where $N_S(t_0, t)$ denote the number of S-interval in time interval $[t_0, t)$ and $N_D(t_0, t)$ denote the number of D-interval in time interval $[t_0, t)$.

Generally, the DoS attacks are sneaking and energy-limited. Without loss of generality, we firstly give the following assumption.

Assumption 1. *For a sneaking DoS attack with limited attack energy, the maximum continuous number of data losses induced by DoS attacks is equal or smaller than M . Here, $M \geq 0$ is an integer.*

Remark 1. *Different from those in [10], [25], they are required that the DoS attacks satisfy the periodic or probabilistic distribution, In Assumption 1, only a constraint on DoS attack energy is needed. This is more practical in a sneaking attack.*

Since there is DoS attack in the communication, the triggered data in $\mathcal{S}_2$ can not always arrive at the controller, that is, the event-triggered state $x(s_k)$ in (3) cannot be used in (4) again. How to deal with this case will be concerned in the following Section III.

III. SWITCHING-LIKE EVENT-TRIGGERED COMMUNICATION SCHEME

The purpose of the section is to propose a switching-like event-triggered communication scheme for NCSs under DoS attacks described in Section II. As shown in Fig. 1, The ACK technique is used to detect whether or not there are data dropouts induced by DoS attacks at every event-triggered transmission instant.

Now we are in a position to design a switching-like event-triggered communication scheme. As described in Fig. 2, at the current event-triggered communication instant, if $ACK = 1$, this means $t \in \mathcal{I}_{t_k}$ and there is no DoS attack at the current event-triggered communication instant, then the event-triggered communication parameter in (2) is set as δ_1 ; otherwise, $ACK = 0$, this implies $t \in \bar{\mathcal{I}}_{t_k}$, the event-triggered communication parameter in (2) is set as δ_2 , where $\mathcal{I}_{t_k}$ and $\bar{\mathcal{I}}_{t_k}$ are given in Definition 1. The event-triggered communication parameter is switched between δ_1 and δ_2 , which is decided by the ACK detective signal.

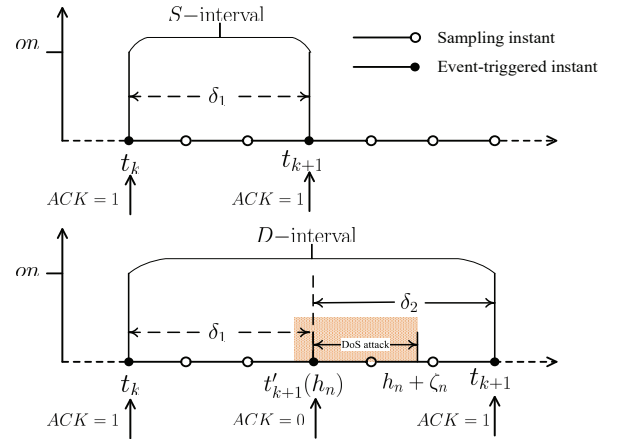


Fig. 2. Switching of event-triggered scheme under DoS attacks

The above mentioned switching-like event-triggered communication scheme can be described as follows.

$$t_{k+1} = t_k + \begin{cases} \inf\{\ell h | e^T(i_k h) \Phi e(i_k h) \geq \delta_1 x(t_k)^T \Phi x(t_k)\}, & ACK = 1 \\ \inf\{\ell h | e^T(i_k h) \Phi e(i_k h) \geq \delta_2 x(t_k)^T \Phi x(t_k)\}, & ACK = 0 \end{cases} \quad (7)$$

where $i_k h = t_k + \ell h$, $\ell = 1, 2, \dots$, $e(i_k h)$ is defined in (2), δ_1 and δ_2 are event-triggered communication parameters and Φ is a positive definite matrix to be designed.

Notice that if the ACK packet from the controller to the sensor is also lost due to the DoS attacks in Fig. 1, in this case, the ACK is set as 0 and the event-triggered communication parameter in (7) is chosen as δ_2 . As a special case, if the ACK packet is always lost, then the switching-like event-triggered communication scheme (7) is simplified as the event-triggered scheme in [16].

For further analysis, we divide the holding interval of the ZOH into the following sampling-interval-like subsets $\Omega_l = [i_k h + \tau_{i_k}, i_k h + h + \tau_{i_{k+1}}]$ with $l = 0, \dots, t_{k+1} - t_k - h$, and define $\eta(t) = t - i_k h$, $t \in \Omega_l$. It is easy to see that $\eta(t)$ is a

piece-wise function satisfying $0 < \eta_1 \leq \eta(t) \leq \eta_2 + h$, where η_1 and η_2 are the minimum and maximum communication delay for τ_{ik} , respectively. Furthermore, it is not necessary that $\eta_2 < h$.

Then, by combining (4) and (7), the closed-loop NCS model can be given by

$$\begin{cases} \dot{x}(t) = Ax(t) + BK(x(t - \eta(t)) - e(i_k h)) + B_w w(t) \\ z(t) = Cx(t) + DK(x(t - \eta(t)) - e(i_k h)) \end{cases} \quad (8)$$

where the initial condition of $x(t)$ is given by $\phi(t_0) = x(t_0)$ and $\phi(t)$ is a continuous function on $[t_0 - \eta_2 + h, t_0]$.

By considering the division of communication intervals, two event-triggered schemes in (7) will be switched between S-interval $\mathcal{I}_{t_k}$ and D-interval $\bar{\mathcal{I}}_{t_k}$. How to estimate the allowable number of packet dropouts for the system (8) under the event-triggered communication scheme (7) will be discussed in the following section.

The purpose of this paper is to find the switching-like event triggering parameters δ_1 , δ_2 and design a security controller gain K to stabilize the NCS (8) with DoS attacks while achieving the following control objectives

- The system (8) is asymptotically stable when there is no disturbance, i.e. $w(t) = 0$;
- A prescribed H_∞ performance $\|z(t)\|_2 \leq \gamma \|w(t)\|_2$ is reachable for any nonzero $w(t) \neq 0$ under zero initial condition.

Remark 2. Different from the adaptive event-triggered scheme in [26], which determines the event-triggered parameter according to the change of system state, the selection of the event-triggered parameter for the proposed switching-like event-triggered scheme is determined by the detection of DoS attacks. Moreover, compared with the resilient event-triggered scheme in [16], where a smaller event-triggered parameter is adopted to alleviate the influence of DoS attacks, in this work, since the event-triggered communication parameter is switched based on whether there are DoS attacks or not, higher communication efficiency can be expected, which will be verified in the numerical example section.

IV. MAIN RESULTS

In this section, the stability analysis of NCSs under the switching-like event-triggered communication scheme is firstly conducted in Theorem 1. Then, based on Theorem 1, how to design the H_∞ controller gain for the studied system is addressed in Theorem 2.

Before proceeding further, the following lemma is given to show that the general stability criterion with event-triggered communication (2) and without DoS attacks.

Lemma 1. For some given positive constants, η_1 , $\eta_2 (\geq \eta_1)$, γ , and a controller gain matrix K , the studied NCS (8) under the event-triggered communication scheme (2) is asymptotically stable with an H_∞ performance index γ , if the following inequalities hold with the real matrices $P > 0$, $R > 0$, $\Phi > 0$, $Q_i > 0$ ($i = 1, 2$), S of appropriate dimensions,

$$\begin{bmatrix} \Xi_{11} & * \\ \Xi_{21} & \Xi_{22} \end{bmatrix} < 0, \begin{bmatrix} Z_2 & * \\ S & Z_2 \end{bmatrix} > 0 \quad (9)$$

where $\Xi_{11} =$

$$\begin{aligned} &[(1, 1) = A^T P + P A + Q_1 - Z_1 - \frac{\pi^2}{4} R, \\ &(2, 1) = Z_1, \\ &(2, 2) = Q_2 - Q_1 - Z_1 - Z_2, \\ &(3, 1) = (PBK)^T + \frac{\pi^2}{4} R + \delta_1 \Phi, \\ &(3, 2) = Z_2 - S, \\ &(3, 3) = -2Z_2 + S + S^T - \frac{\pi^2}{4} R + \delta_1 \Phi, \\ &(4, 2) = S, \\ &(4, 3) = Z_2 - S, \\ &(4, 4) = -Z_2 - Q_2, \\ &(5, 1) = -(PBK)^T, \\ &(5, 3) = -\delta_1 \Phi, \\ &(5, 5) = -\Phi + \delta_1 \Phi, \\ &(6, 1) = (PB_w)^T, \\ &(6, 6) = -\gamma^2 I], \\ &\Xi_{21} = \text{col}\{\eta_1 Z_1 \Gamma_1, (\eta_2 - \eta_1) Z_2 \Gamma_1, \eta_2 R \Gamma_1, \Gamma_2\}, \\ &\Xi_{22} = \text{diag}\{-Z_1, -Z_2, -R, -I\}, \\ &\text{with} \\ &\Gamma_1 = [A, 0, BK, 0, -BK, B_w], \\ &\Gamma_2 = [C, 0, DK, 0, -DK, 0]. \end{aligned}$$

Proof: See Appendix.

When there are DoS attacks in the communication, based on Lemma 1, one can derive the following Theorem to 1) judge the stability of the studied system (8) under the switching-like event-triggered communication scheme (7); and 2) evaluate the event-triggered parameter δ_2 in (7).

Theorem 1. For some given positive constants, η_1 , $\eta_2 (\geq \eta_1)$, γ , and a controller gain matrix K , under the switching-like event-triggered scheme (7), the system (8) is asymptotically stable with an H_∞ performance index γ , if the inequalities (9) hold with the real matrices $P > 0$, $R > 0$, $\Phi > 0$, $Q_i > 0$ ($i = 1, 2$), S of appropriate dimensions, and δ_2 satisfies

$$\delta_2 \leq \left({}^{M+1}\sqrt{\frac{1 + \sqrt{\delta_1}}{(1 + \varepsilon)^M}} - 1 \right)^2 \quad (10)$$

where M is the allowable maximum number of data dropouts induced by DoS attacks in Assumption 1, δ_1 and δ_2 are given in (7), and $\varepsilon = |Ah| + |\frac{hBK}{1 - \sqrt{\delta_1}}| + |\nu h B_w|$, and ν is a given positive constrained constant.

Proof: See Appendix.

Remark 3. From the proof of Theorem 1, one can see that if there are maximal M continuous packets losses in the communication due to DoS attacks, the proposed first event-triggered condition in (7) can guarantee that the second event-triggered condition in (7), therefore, Theorem 1 can be derived based on Lemma 1. In other words, if there are M continuous packet dropouts in the communications, the desired control performance can be guaranteed by using the proposed event-triggered scheme (7).

Based on Theorem 1, the following Theorem 2 can be derived to design the H_∞ security controller gain.

Theorem 2. If there exist positive scalars η_1 , η_2 , δ_1 , γ , real matrices $\tilde{X} > 0$, $\tilde{R} > 0$, $\tilde{\Phi} > 0$, $\tilde{Q}_i > 0$ ($i = 1, 2$) and $\tilde{S}$ of

appropriate dimensions, and δ_2 satisfies (10), such that

$$\begin{bmatrix} \tilde{\Xi}_{11} & * \\ \tilde{\Xi}_{21} & \tilde{\Xi}_{22} \end{bmatrix} < 0, \begin{bmatrix} \tilde{Z}_2 & * \\ \tilde{S} & \tilde{Z}_2 \end{bmatrix} > 0 \quad (11)$$

where $\tilde{\Xi}_{11} =$

$$[(1,1) = XA^T + AX + \tilde{Q}_1 - \tilde{Z}_1 - \frac{\pi^2}{4}\tilde{R},$$

$$(2,1) = \tilde{Z}_1,$$

$$(2,2) = \tilde{Q}_2 - \tilde{Q}_1 - \tilde{Z}_1 - \tilde{Z}_2,$$

$$(3,1) = (BY)^T + \frac{\pi^2}{4}\tilde{R} + \delta_1\tilde{\Phi},$$

$$(3,2) = \tilde{Z}_2 - \tilde{S},$$

$$(3,3) = -2\tilde{Z}_2 + \tilde{S} + \tilde{S}^T - \frac{\pi^2}{4}\tilde{R} + \delta_1\tilde{\Phi},$$

$$(4,2) = \tilde{S},$$

$$(4,3) = \tilde{Z}_2 - \tilde{S},$$

$$(4,4) = -\tilde{Z}_2 - \tilde{Q}_2,$$

$$(5,1) = -(BY)^T,$$

$$(5,3) = -\delta_1\tilde{\Phi},$$

$$(5,5) = -\tilde{\Phi} + \delta_1\tilde{\Phi},$$

$$(6,1) = (B_w)^T,$$

$$(6,6) = -\gamma^2 I],$$

$$\tilde{\Xi}_{21} = \text{col}\{\eta_1 \tilde{Z}_1 \Gamma_1, (\eta_2 - \eta_1) \tilde{Z}_2 \Gamma_1, \eta_2 \tilde{R} \Gamma_1, \Gamma_2\},$$

$$\tilde{\Xi}_{22} = \text{diag}\{-\tilde{Z}_1, -\tilde{Z}_2, -\tilde{R}, -I\},$$

with

$$\Gamma_1 = [AX, 0, BY, 0, -BY, B_w],$$

$$\Gamma_2 = [CX, 0, DY, 0, -DY, 0].$$

Then under the switching-like event-triggered scheme (7), the NCS (8) is asymptotically stable with controller gain $K = YX^{-1}$ and an H_∞ performance index γ .

Proof: Define $X = P^{-1}$, $\tilde{R} = XRX$, $\tilde{Q}_i = XQ_iX$, $\tilde{Z}_i = XZ_iX$ ($i = 1, 2$), $\tilde{S} = XSX$, $\tilde{\Phi} = X\Phi X$ and $Y = KX$. Then pre- and post- multiplying both sides of the left term of inequality with $\text{diag}[X, X]$ and right term of inequality with $\text{diag}[X, X, X, X, X, I, X, X, X, I]$ for (11), we can arrive at the desired results in Theorem 2 by using the fact that $-HG^{-1}H \leq G - 2H$ for appropriate matrices to deal with the non-linear terms. This completes the proof. ■

Remark 4. From Theorem 2, one can see that the security controller gain $K = YX^{-1}$, which is related to the event-triggered parameters δ_1 , δ_2 , M , and control performance γ . If the maximum allowable number of data dropouts M and δ_1 are known, based on Theorem 1, one can derive the δ_2 from (10); if δ_1 , δ_2 are pre-given in (7), one can also derive that M from (10). The similar trade-off algorithm in [19] can also be used to choose the parameters δ_1 , δ_2 , K under the constraints of maximum allowable number of data dropouts M and H_∞ control performance index γ .

V. AN ILLUSTRATED EXAMPLE

In this section, an example is used to verify the effectiveness of proposed method. Consider the inverted pendulum on a cart borrowed from [19], the networked control structure is shown in Fig. 3.

Suppose that the dynamics of the controlled plant are given

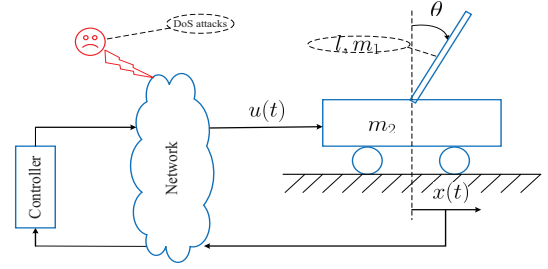


Fig. 3. Networked pendulum control on a cart

by

$$\dot{x}(t) = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & -\frac{m_1 g}{m_2} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -\frac{g}{l} & 0 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ \frac{1}{m_2} \\ 0 \\ -\frac{1}{m_2 l} \end{bmatrix} u(t) + \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} w(t) \quad (12)$$

where $m = 1$ is the mass of the pendulum bob, $m_2 = 10$ is the cart mass, $l = 3$ is the length of the pendulum arm and $g = 10$ is the gravitational acceleration. The initial state is $x(0) = [0.98 \ 0 \ 0.2 \ 0]^T$. One can see that the system is unstable when there are no control input. Set $\gamma = 200$, $\eta_1 = 0$, $\eta_2 = 0.001$, $w(t) = 0.001 \sin(t)$ and simulation time $[0, 30s]$ with sampling period $h = 0.001s$.

Case I: DoS-free case with $\delta_1 = 0.2$

According to Theorem 2, one can design a controller under event-triggered communication scheme with $\delta_1 = 0.2$. By solving LMIs, the corresponding feedback controller gain can be obtained as

$$K_1 = [2.8463 \ 10.9743 \ 304.5925 \ 169.6811] \quad (13)$$

The state response and release instants without DoS attacks are shown in Fig. 4, where the number of triggered packets to be transmitted is 150. Obviously, the stability of the studied (12) is verified.

Case II: DoS case with $\delta_1 = 0.2$

Suppose that the maximum allowable successive packet dropouts $M = 2$ due to DoS attacks, then the simulation on the state response under DoS-introduced packet dropouts is shown in Fig. 5. From Fig. 5, one can see that the stability is

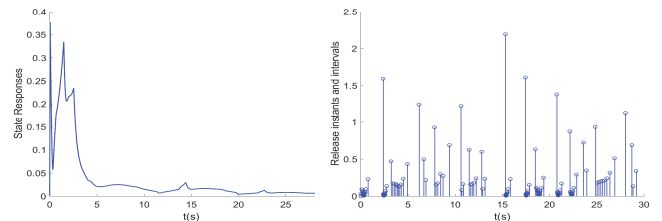


Fig. 4. State response and release instants without DoS attacks

lost when there are DoS attacks. This implies that control with the event-triggered communication scheme (2) is vulnerable to DoS attacks.

Case III: Resilient event-triggered control scheme with $\delta_2 = 0.00036$

The simulation on the resilient event-triggered scheme proposed in [16] is given for the transmission comparison. With

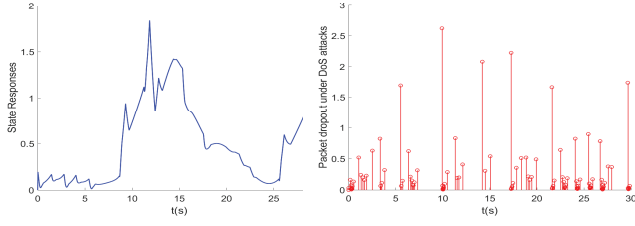


Fig. 5. State response under DoS attacks in Case II

$M = 2$, one can design the event-triggered communication parameter $\delta_2 = 0.00036$ to deal with packet dropouts and the controller gain is given as

$$K_2 = \begin{bmatrix} 8.6380 & 21.3446 & 369.1400 & 207.3368 \end{bmatrix} \quad (14)$$

according to Theorem 2 in [16]. The state response and the release instants are shown in Fig. 6.

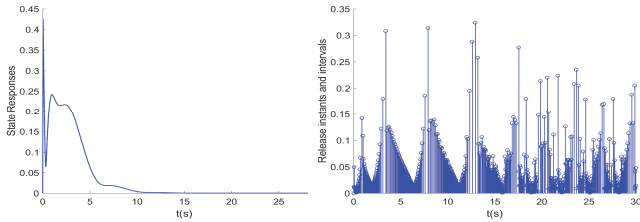


Fig. 6. State response and release instants in Case III

Compared with the state responses in Fig. 5 and Fig. 6, one can see that the stability of the studied system is kept in this case, and larger number of packets, i.e., 1083 has been transmitted to deal with DoS attacks, which is more than 150 in Case I.

Case IV: Switching-like event-triggered scheme (7)

By using the ACK detection mechanism, the switching-like event-triggered scheme (7) is employed to save the limited communication resources while keeping the desired control performance. Suppose that the DoS attacks probability is 0.8. Namely, $\delta_1 = 0.2$ when $ACK = 1$, $\delta_2 = 0.00036$ when $ACK = 0$.

The state response and the release interval by using switching-like event-triggered communication scheme (7) are shown in Fig. 7, which shows that there are only 594 trans-

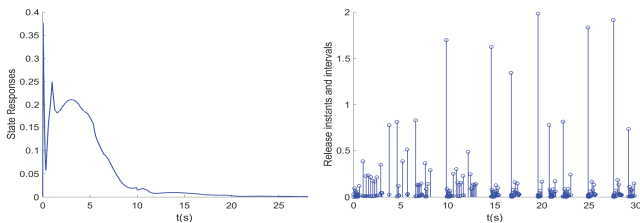


Fig. 7. State response and release interval under switching-like event-triggered scheme (7)

mitted packets, it is less than 1083 in Case III. Moreover, the stability of the studied system (12) is also guaranteed.

For comparison, the number of transmitted packets and stability under different event-triggered schemes are listed in Table I.

TABLE I
EVENT-TRIGGERED PARAMETERS AND STABILITY OF THE STUDIED SYSTEM UNDER DIFFERENT EVENT-TRIGGERED SCHEMES

Methods	δ_1	δ_2	Gain	Transmitted packets	Stability
ETC in [19]	0.2	/	K_1	150	No
ETC in [16]	/	0.00036	K_2	1083	Yes
This paper	0.2	0.00036	K_1	594	Yes

From Table I, one can see that

- the event-triggered scheme in [19] is vulnerable to DoS attacks. The stability of the studied system is not kept when there is a DoS attack;
- the resilient event-triggered scheme in [16] is robust to DoS attacks. Compared with the number of transmitted packets in [19], more packets must be delivered to deal with potential DoS attacks.
- compared with the resilient event-triggered schemes in [16], the switching-like event-triggered scheme proposed in this paper can save more communication resources while preserving the desired control performance.

VI. CONCLUSION

In this paper, a switching-like event-triggered control method has been proposed, which integrates communication parameters, DoS characteristics and controller design in a unified framework. In comparison with the existing resilient event-triggered communication scheme, by using the proposed switching-like event-triggered communication scheme, a higher communication efficiency has been obtained under the DoS attacks while ensuring the desired control performance. How to optimally design the event-triggered communication schemes under different DoS attack frequency is left for our further study.

VII. APPENDIX

A. Proof of Lemma 1

Consider the following candidate Lyapunov-Krasovskii functional $V(t)$,

$$V(t) = V_1(t) + V_2(t) + V_3(t) + V_4(t) \quad (15)$$

where

$$V_1(t) = x^T(t)Px(t)$$

$$V_2(t) = \int_{t-\eta_1}^t x^T(s)Q_1x(s)ds + \int_{t-\eta_2}^{t-\eta_1} x^T(s)Q_2x(s)ds$$

$$V_3(t) = \eta_1 \int_{-\eta_1}^0 \int_{t+\theta}^t \dot{x}^T(s)Z_1\dot{x}(s)dsd\theta + (\eta_2 - \eta_1) \int_{-\eta_2}^{-\eta_1} \int_{t+\theta}^t \dot{x}^T(s)Z_2\dot{x}(s)dsd\theta$$

$$V_4(t) = \eta_2^2 \int_{i_k h}^t \dot{x}^T(s)R\dot{x}(s)ds - \frac{\pi^2}{4} \int_{i_k h}^t [x(s) - x(i_k h)]^T R [x(s) - x(i_k h)] ds$$

Take the time derivative along the trajectory of the system (8) yields

$$\begin{aligned}
\dot{V}_1(t) &= 2x^T(t)P[Ax(t) + BKx(t - \eta(t)) - BKe(i_k h) \\
&\quad + B_w w(t)] \\
\dot{V}_2(t) &= x^T(t)Q_1x(t) - x^T(t - \eta_2)Q_2x(t - \eta_2) + \\
&\quad x^T(t - \eta_1)(Q_2 - Q_1)x(t - \eta_1) \\
\dot{V}_3(t) &= \dot{x}^T(t)(\eta_1^2 Z_1 + (\eta_2 - \eta_1)^2 Z_2)\dot{x}(t) \\
&\quad - \eta_1 \int_{t-\eta_1}^t \dot{x}^T(s)Z_1\dot{x}(s)ds \\
&\quad - (\eta_2 - \eta_1) \int_{t-\eta_2}^{t-\eta_1} \dot{x}^T(s)Z_2\dot{x}(s)ds \\
\dot{V}_4(t) &= \eta_2^2 \dot{x}^T(t)R\dot{x}(t) \\
&\quad - \frac{\pi^2}{4}[x(t) - x(t - \eta(t))]^T R[x(t) - x(t - \eta(t))]
\end{aligned} \tag{16}$$

By using Jessen's inequality [27] to deal with one cross item in $\dot{V}_3(t)$, we have

$$\begin{aligned}
& -\eta_1 \int_{t-\eta_1}^t \dot{x}^T(s)Z_1\dot{x}(s)ds \leq -\int_{t-\eta_1}^t \dot{x}^T(s)ds Z_1 \int_{t-\eta_1}^t \dot{x}(s)ds \\
& = \begin{bmatrix} x(t) \\ x(t - \eta_1) \end{bmatrix} \begin{bmatrix} Z_1 & -Z_1^T \\ * & Z_1 \end{bmatrix} \begin{bmatrix} x(t) \\ x(t - \eta_1) \end{bmatrix}
\end{aligned} \tag{17}$$

The another cross item in $\dot{V}_3(t)$ can be written as:

$$\begin{aligned}
& (\eta_2 - \eta_1) \int_{t-\eta_2}^{t-\eta_1} \dot{x}^T(s)Z_2\dot{x}(s)ds = \\
& (\eta_2 - \eta_1) \left[\int_{t-\eta(t)}^{t-\eta_1} \dot{x}^T(s)Z_2\dot{x}(s)ds + \int_{t-\eta_2}^{t-\eta(t)} \dot{x}^T(s)Z_2\dot{x}(s)ds \right]
\end{aligned} \tag{18}$$

Then, by using reciprocally convex approach [28] to deal with two cross items in (18), if there exist $\begin{bmatrix} Z_2 & S^T \\ * & Z_2 \end{bmatrix} > 0$, we have

$$\begin{aligned}
& -(\eta_2 - \eta_1) \int_{t-\eta_2}^{t-\eta_1} \dot{x}^T(s)Z_2\dot{x}(s)ds \leq \\
& -\varsigma^T(t) \begin{bmatrix} Z_2 & S^T - Z_2^T & -S^T \\ * & 2Z_2 - S - S^T & S^T - Z_2^T \\ * & * & Z_2 \end{bmatrix} \varsigma(t)
\end{aligned} \tag{19}$$

with $\varsigma(t) = [x^T(t - \eta_1) \quad x^T(t - \eta(t)) \quad x^T(t - \eta_2)]^T$.

Furthermore, taking the first event-triggered scheme in (7) and $z^T(t)z(t) \leq \gamma^2 w^T(t)w(t)$ into consideration, the following inequality holds

$$\dot{V}(t) \leq \xi^T(t)\Xi\xi(t) - z^T(t)z(t) + \gamma^2 w^T(t)w(t) \tag{20}$$

where $\xi(t) = [x(t), x(t - \eta_1), x(t - \eta(t)), x(t - \eta_2), e(i_k h), w(t)]$, and $\Xi = \Xi_{11} + \Gamma_1^T[\eta_1 Z_1 + (\eta_2 - \eta_1)Z_2 + \eta_2 R]\Gamma_1 + \Gamma_2^T \Gamma_2$.

Then using Schur complement with (20), one can easily obtain that (9) holds. Recalling $\Xi < 0$ and $\dot{V}(t)$ is continuous in t , the

integration of both sides of $\dot{V}(t) \leq -z^T(t)z(t) + \gamma^2 w^T(t)w(t)$ form 0 to $+\infty$ leads to

$$V(+\infty) - V(0) < \int_0^{+\infty} [-z^T(t)z(t) + \gamma^2 w^T(t)w(t)]dt \tag{21}$$

Therefore, it is clear that there exists a sufficiently small positive scalar ε such that $V(t) \leq \varepsilon \|x(t)\|$ when (9) holds. This completes the proof. ■

B. Proof of Theorem 1

Based on Lemma 1, the stability of studied NCS (8) with switching-like event-triggered communication scheme (7) and H_∞ performance is proved by the following two cases.

- 1) If there are no DoS attacks, the first event-triggered condition with δ_1 is adopted in (7). Based on the proof of Lemma 1, one can easily obtain that the studied NCS (8) is stable with the desired H_∞ performance.
- 2) For the DoS case, similar to [19], suppose that there are successive M packet dropouts with $t_k = t_0 < d_1 < d_2 < \dots < d_l < \dots < d_M < d_{M+1} = t_{k+1}$ under the event-triggered parameter δ_2 . Here, $(l = 0, 1, \dots, M)$. Applying the first event-triggered conditions in (7) yields

$$|x(t_k)| \leq \sqrt{\delta_1}|x(t_k)| + |x(d_{l+1} - h)| \tag{22}$$

and

$$(1 - \sqrt{\delta_1})|x(t_k)| \leq |x(d_{l+1} - h)| \tag{23}$$

Applying the second event-triggered condition in (7) yields

$$|x(d_{l+1} - h) - x(d_l)| \leq \sqrt{\delta_2}|x(d_l)| \tag{24}$$

and

$$\begin{aligned}
|x(d_{l+1} - h)| &\leq |x(d_{l+1} - h) - x(d_l)| + |x(d_l)| \\
&\leq (1 + \sqrt{\delta_2})|x(d_l)|
\end{aligned} \tag{25}$$

For the external disturbance $w(t)$, we assume that $|w(t)| \leq \nu|x(t)|$ for all $t \geq 0$. According to the solution of $x(t)|_{t=d_{l+1}h}$, then

$$\begin{aligned}
& |x(d_{l+1}) - x(d_{l+1} - h)| \\
& \leq |Ahx(d_{l+1} - h) + |hBx(t_k)| \\
& \quad + |hB_w w(d_{l+1} - h)| \\
& \leq \varepsilon|x(d_{l+1} - h)|
\end{aligned} \tag{26}$$

where ε is given in Theorem 1.

Recalling the fact that (25) and (26), we have

$$|x(d_{l+1}) - x(d_{l+1} - h)| \leq \varepsilon(1 + \sqrt{\delta_2})|x(d_l)| \tag{27}$$

Combine (22) and (27), the error between t and $t_k h$ is

$$\begin{aligned}
& |x(t) - x(t_k)| \\
& \leq \sum_{l=0}^M \sqrt{\delta_2}|x(d_l)| + \sum_{l=0}^{M-1} \varepsilon(1 + \sqrt{\delta_2})|x(d_l)|
\end{aligned} \tag{28}$$

From (24) and (26), one can obtain

$$\begin{aligned}
& |x(d_{l+1})| \leq |x(d_{l+1}) - x(d_{l+1} - h)| \\
& \quad + |x(d_{l+1} - h) - x(d_l)| + |x(d_l)| \\
& \quad [(1 + \varepsilon)(1 + \sqrt{\delta_2})]^{l+1}|x(t_k)|
\end{aligned} \tag{29}$$

From (28) and (29), one can obtain that

$$\begin{aligned} & |x(t) - x(t_k h)| \\ & \leq ((1 + \sqrt{\delta_2})^{M+1} (1 + \varepsilon)^M - 1) |x(t_k)| \end{aligned} \quad (30)$$

Considering the first event-triggered scheme in (7) and (30) together leads to

$$(1 + \sqrt{\delta_2})^{M+1} (1 + \varepsilon)^M - 1 \leq \sqrt{\delta_1} \quad (31)$$

From (31), it is clear to see that the first event-triggered condition can be derived by the second one under the successive M packet dropouts. Then, the stability of studied NCS (8) with H_∞ performance can be achieved based on the proof of Lemma 1.

Combing above two cases, the stability of the studied NCS (8) with H_∞ performance under the DoS-introduced switching-like event-triggered scheme (7) is proved. ■

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